## 1.2 Row reduction and Echelon forms

A nonzero row or column in a matrix means a row or column that contains at least one nonzero entry, a leading entry of a row refers to the leftmost nonzero entry.

The following matrices are in echelon form. The leading entries (■) may have any nonzero value; the starred entries (*) may have any value (including zero).

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

The following matrices are in reduced echelon form because the leading entries are 1's, and there are 0's below *and above* each leading 1.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

A pivot position in a matrix $A$ is a location in $A$ that corresponds to a leading 1 in the reduced echelon form of $A$. A pivot column is a column of $A$ that contains a pivot position.

## Row Reduction Algorithm

Example: Apply elementary row operations to transform the following matrix into echelon form and then into reduced echelon form.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

Step 1: Begin with the leftmost nonzero column. This is a pivot column. The pivot position is at the top.

$$\begin{bmatrix} 0 & 3 & -6 & 6 & 4 & -5 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 3 & -9 & 12 & -9 & 6 & 15 \end{bmatrix}$$

pivot column

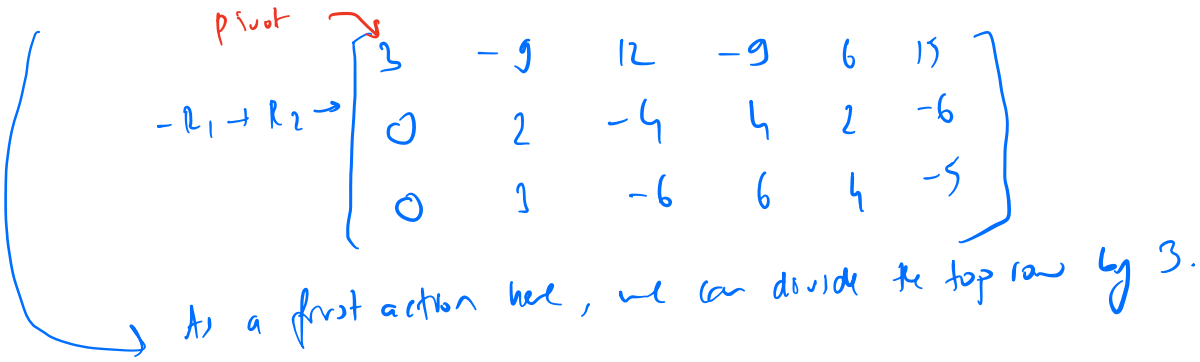
Step 2: Select a nonzero entry in the pivot column as a pivot. If necessary, interchange rows to move this entry into the pivot position.

Interchange rows 1 and 3. (interchanging rows 1 and 2 is another possibility)

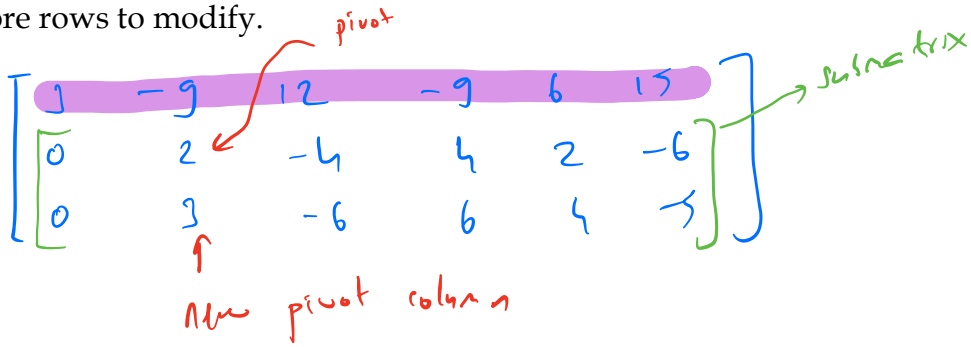
pivot

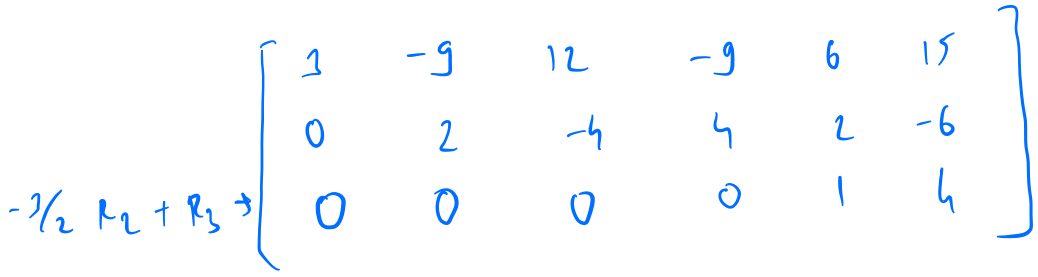
$$\begin{bmatrix} 3 & -9 & 12 & -9 & 6 & 15 \\ 3 & -7 & 8 & -5 & 8 & 9 \\ 0 & 3 & -6 & 6 & 4 & -5 \end{bmatrix}$$

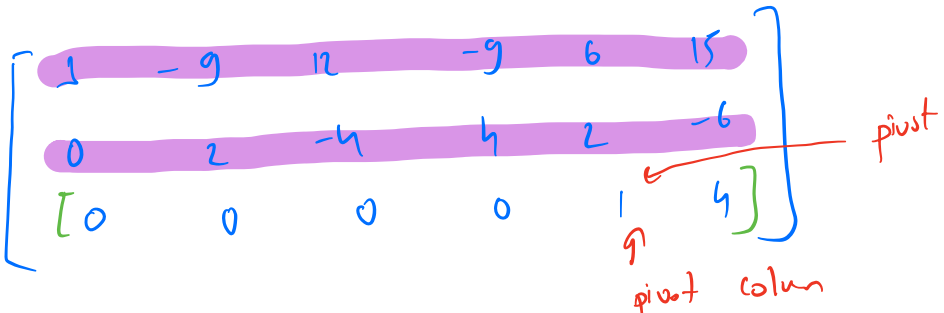
Step 3: Use row replacement operations to create zeros in all positions below the pivot.




Step 4: Cover (or ignore) the row containing the pivot position and cover all rows, if any, above it. Apply steps 1-~~3~~ 3 to the submatrix that remains. Repeat the process until there are no more rows to modify.

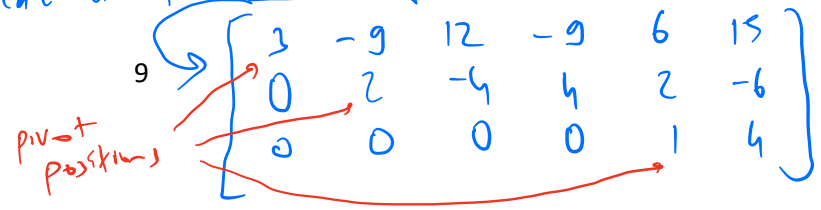



Step 1-3 require no work for this submatrix.

Then we have reached the <u>echelon form</u> of the full matrix.



Step 5: Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot. If a pivot is not 1 by a scaling operation.

make it 1

$$\begin{bmatrix} 1 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -9 & 12 & -9 & 0 & -9 \\ 0 & 2 & -4 & 4 & 0 & -14 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \begin{matrix} \leftarrow R_1 - 6R_3 \\ \leftarrow R_2 - 2R_3 \end{matrix}$$

$$\begin{bmatrix} 1 & -9 & 12 & -9 & 0 & -9 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow R_2/2$$

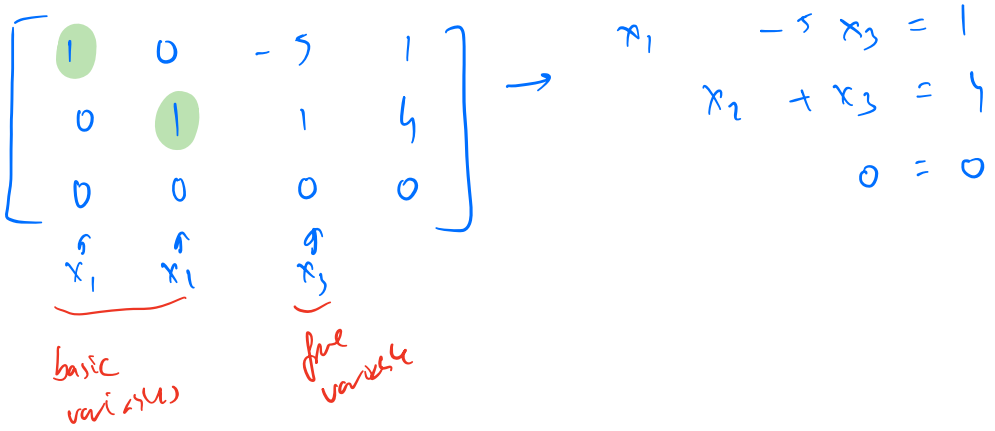
$$\begin{bmatrix} 1 & 0 & -6 & 9 & 0 & -72 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow 9R_2 + R_1$$

$$\begin{bmatrix} 1 & 0 & -2 & 3 & 0 & -44 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{bmatrix} \leftarrow R_1/3$$

This is the reduced echelon form.

Solutions of linear systems

Suppose that the augmented matrix of a linear system has been changed into the equivalent reduced echelon form.




The variables $x_1$ and $x_2$ corresponding to pivot columns in the matrix are called basic variables. The other variable, $x_3$, is called a free variable.

Whenever a system is consistent the solution set can be described explicitly by solving the reduced system of equations for the basic variables in terms of the free variables. Then,

$x_1 = 1 + 5x_3$

$x_2 = 4 - x_3$

$x_3$ is free → you are free to choose any value for $x_3$.

If $x_3 = 0$, then $x_1 = 1$, $x_2 = 4$

$x_3 = 1$, " $x_1 = 6$, $x_2 = 3$

Find the general solution of the system whose augmented matrix is reduced to:

Ex:

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{2R_1+R_2} \xrightarrow{R_3+R_2} \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

pivot

$$\xrightarrow{R_2/2} \begin{bmatrix} 1 & 6 & 2 & -5 & 0 & 10 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{R_1-2R_2} \begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

pivot

$x_1, x_3, x_5$ are basic

$x_2, x_4$ are free.

$x_1 + 6x_2 + 1x_4 = 0$

$x_3 - 4x_4 = 5$

$x_5 = 7$

Then, the general solution is:

$x_1 = -6x_2 - 3x_4$

$x_2$ is free

$x_3 = 5 + 4x_4$

$x_4$ is free

$x_5 = 7$

Whenever a system is consistent and has free variables, the solution set has many parametric descriptions. In the previous example, we could treat $x_3$ and $x_1$ as a parameter, we would have an accurate description of the solution set. However, to be consistent, we always use the free variables as the parameters for describing a solution set.

Existence and uniqueness theorem

**Existence and Uniqueness Theorem**

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$[0 \quad \cdots \quad 0 \quad b] \qquad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.